

# MA388 Sabermetrics: Lesson 13

## Catcher Framing Ability - Visualization

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### Lesson Objectives

1. Import and merge Statcast pitch data with player ID maps.
2. Create faceted visualizations to compare called-strike proportions across catchers.
3. Discuss challenges in isolating catcher framing ability from confounding factors.
4. Calculate called-strike proportions for pitches near the edge of the strike zone.

### Chapter 7

In this chapter, we investigate whether catchers have the ability to consistently frame close pitches, which some believe makes it more likely the umpire will call a strike.

*Do you think some catchers are better than others at framing pitches, and how important is this skill?*

- [Video 1](#)
- [Video 2](#)

*How could we investigate whether some catchers are better than others? List some challenges with answering this question with data.*

```
library(tidyverse)
library(knitr)
library(baseballr)
```

The code below imports the all pitches thrown in May 2019.

```
# If you've already collected the pitch data below, just load it. Otherwise,
# collect it from scratch.

if("statcast_may_2019.rds" %in% dir(getwd())){
  pitches <- readRDS("statcast_may_2019.rds")
}else{

  # Retrieve pitch-level data from May 2019.

  get_statcast_pitches <- function(start_day, end_day, chunk_size = 5) {

    # Coerce to Date
    start_day <- as.Date(start_day)
    end_day   <- as.Date(end_day)

    # Create sequence of chunk start dates
    chunk_starts <- seq(start_day, end_day, by = paste(chunk_size, "days"))

    # Build data
    pitch_data <- map_dfr(chunk_starts, function(chunk_start) {

      chunk_end <- min(chunk_start + days(chunk_size - 1), end_day)
      statcast_search(
        start_date = chunk_start,
        end_date   = chunk_end
      )
    })
    return(pitch_data)
  }

  # If we want to limit to a certain number of pitches, we might take the head()
  # or sample_n() to get the number we want.
  pitches <- get_statcast_pitches('2019-05-01', '2019-05-31')

  saveRDS(pitches, "statcast_may_2019.rds")
}
```

Here is a small sample of the Statcast pitches and variables we've collected.

```
pitches %>%
  select(description, release_speed, pitch_name,
         plate_x, plate_z, balls, strikes, fielder_2) %>%
  head(10) %>%
  kable()
```

|               | re-         |            |         |         |       |         |           |
|---------------|-------------|------------|---------|---------|-------|---------|-----------|
| description   | lease_speed | pitch_name | plate_x | plate_z | balls | strikes | fielder_2 |
| swing-        | 79.2        | Changeup   | -0.54   | 1.79    | 2     | 2       | 444489    |
| ing_strike    |             |            |         |         |       |         |           |
| foul          | 89.3        | Sinker     | -0.74   | 2.42    | 2     | 2       | 444489    |
| foul          | 79.5        | Changeup   | -0.32   | 1.25    | 2     | 1       | 444489    |
| ball          | 88.3        | Sinker     | -1.28   | 1.95    | 1     | 1       | 444489    |
| ball          | 88.1        | Sinker     | -1.82   | 2.29    | 0     | 1       | 444489    |
| called_strike | 87.5        | Sinker     | -0.17   | 2.11    | 0     | 0       | 444489    |
| hit_into_play | 78.6        | Changeup   | -0.27   | 1.20    | 1     | 2       | 444489    |
| ball          | 79.0        | Changeup   | -1.25   | 1.23    | 0     | 2       | 444489    |
| foul          | 78.5        | Changeup   | 0.49    | 1.47    | 0     | 1       | 444489    |
| foul          | 86.6        | Sinker     | 0.38    | 2.52    | 0     | 0       | 444489    |

One of the variables in `pitches` is the catcher's ID. Let's add the catcher's name to each pitch in our data frame using the master list available at <https://www.smartfantasybaseball.com/tools/>.

```
# This CSV is available at https://www.smartfantasybaseball.com/tools/.
mlb_IDs <- read_csv("./PLAYERIDMAP.csv")

pitches <- pitches %>%
  left_join(select(mlb_IDs, PLAYERNAME, MLBID),
           by = c("fielder_2" = "MLBID")) %>%
  rename(catcher_name = PLAYERNAME)
```

Let's look at the 10 catchers with the most pitches thrown to them in early May 2019.

```
top10catchers <- pitches %>%
  count(catcher_name) %>%
  arrange(-n) %>%
  head(10) %>%
  pull(catcher_name)

top10catchers
```

|      |                   |                     |                     |
|------|-------------------|---------------------|---------------------|
| [1]  | "Yasmani Grandal" | "J.T. Realmuto"     | "Willson Contreras" |
| [4]  | "Yadier Molina"   | "Elias Diaz"        | "Martin Maldonado"  |
| [7]  | "Jorge Alfaro"    | "Robinson Chirinos" | "Wilson Ramos"      |
| [10] | "Josh Phegley"    |                     |                     |

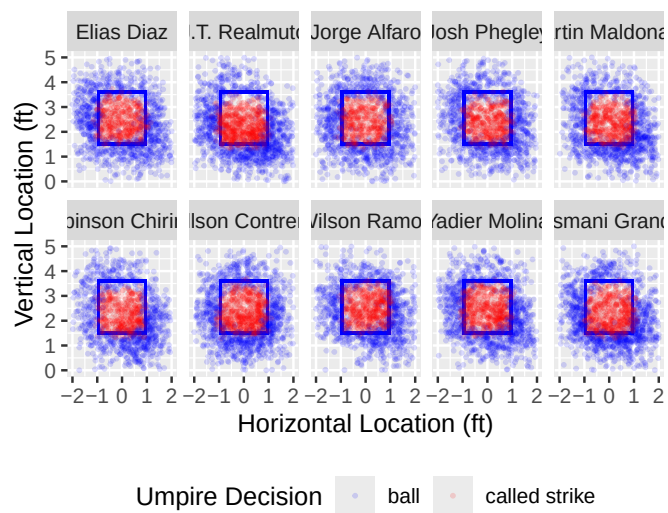
Okay, let's see if we can see anything by looking at called strikes versus balls.

```
# Let's plot the strike zone.
# Note px and pz in the book are actually plate_x and plate_z, respectively.
plate_width <- 17 + 2 * (9/pi)
k_zone_plot <- ggplot(NULL, aes(x = plate_x, y = plate_z)) +
  geom_rect(xmin = -(plate_width / 2) / 12,
            xmax = (plate_width / 2) / 12,
            ymin = 1.5,
            ymax = 3.6,
            color = "blue",
            alpha = 0) +
  coord_equal() +
  scale_x_continuous("Horizontal Location (ft)", limits = c(-2,2)) +
  scale_y_continuous("Vertical Location (ft)", limits = c(0,5))

# Look at only taken pitches caught by one of the 10 top catchers.
pitches_taken <- pitches %>%
  filter(catcher_name %in% top10catchers,
         description %in% c("ball", "called_strike")) %>%
  mutate(description = str_replace(description, "_", " "))

# Plot pitches that were balls or called strikes.
k_zone_plot %+% pitches_taken +
  aes(color = description) +
  geom_point(alpha = 0.15, size = .5) +
  scale_color_manual(values = c("blue", "red")) +
  facet_wrap(~catcher_name, ncol = 5) +
  theme(legend.position = "bottom") +
  labs(title = "Strike vs. Ball Calls on Taken Pitches",
       color = "Umpire Decision")
```

### Strike vs. Ball Calls on Taken Pitches



How can we make some sense of the different panels in this figure? Look closely – does anything jump out at you?

Okay, this is admittedly pretty subjective, and we can't stop here. We need a quantitative assessment of how different catchers may or may not be contributing to different proportions of called strikes on pitches where batters didn't swing.

```
pitches_taken %>%
  count(catcher_name, description) %>%
  group_by(catcher_name) %>%
  mutate(prop = n / sum(n)) %>%
  filter(description == "called strike") %>%
  arrange(-prop) %>%
  kable(digits = 3,
        caption = "Proportion of taken pitches that were called strikes by catcher.")
```

Table 1: Proportion of taken pitches that were called strikes by catcher.

| catcher_name      | description   | n   | prop  |
|-------------------|---------------|-----|-------|
| Wilson Ramos      | called strike | 504 | 0.362 |
| Willson Contreras | called strike | 575 | 0.356 |
| J.T. Realmuto     | called strike | 602 | 0.353 |
| Yadier Molina     | called strike | 533 | 0.346 |
| Martin Maldonado  | called strike | 496 | 0.337 |
| Robinson Chirinos | called strike | 471 | 0.334 |
| Yasmani Grandal   | called strike | 566 | 0.325 |
| Josh Phegley      | called strike | 443 | 0.320 |
| Elias Diaz        | called strike | 466 | 0.317 |
| Jorge Alfaro      | called strike | 427 | 0.302 |

*Does it appear catchers make a difference?*

*Now that we've been talking about this for a while, what new considerations come to mind?*

*How could a model-based approach improve upon this analysis?*

### Exercise

Calculate that same called strike proportion for pitches that are within 2 inches of the edge of the strike zone?

*Hint: Use `case_when()` to establish a new variable that signifies whether a ball is `sz_edge` based on `plate_x`, `plate_z`, `sz_top`, `sz_bottom` and `-.95` to `.95` for plate edge.*